

## Description

**Graduate assistantship focused on population ecology of moose** – we are seeking a highly motivated person to pursue a PhD degree in the Environmental and Life Sciences Graduate Program at Trent University in Peterborough, Ontario, Canada. The graduate student will join a large team implementing a field research project on the factors limiting and regulating moose populations in Ontario. The field project is a multi-year program that entails the GPS radio collaring of moose, American black bears, wolves and white-tailed deer to assess cause-specific mortality of moose and examine predator-prey dynamics. This project is entering its third year and has collected a large amount of data on moose vital rates and movements. We are seeking a student to focus their research on moose population dynamics and the drivers of variation in moose vital rates. Specific research questions will be determined collaboratively between the student and research team. The successful applicant will have considerable responsibility and freedom to formulate and address basic and applied research questions grounded in ecological theory. Field work will be highly varied and will include live trapping and deployment of collars on black bears, wolves and white-tailed deer and helicopter capture of adult and calf moose. Further, the project involves deployment of remotely triggered trail cameras, collection of non-invasive samples for black bear population estimation, vegetation surveys, hunter surveys, aerial population surveys and more. The student will be expected to conduct significant field work for their specific project AND assist in all field aspects of the broader project. The student will be co-supervised by Dr. Joe Northrup and Dr. Brent Patterson both of the Ontario Ministry of Natural Resources & Trent University.

## Requirements

M.Sc. degree in ecology, wildlife biology, or related field is desired, but *exceptional* past experience may be considered in place of a M.Sc. degree. Applicants must meet the minimum entrance requirements for the Environmental & Life Sciences graduate program at Trent University. Desired qualifications include a GPA >3.5 (4.0 scale). A strong background in ecology, demonstrated analytical capabilities, and passion for wildlife research are required. Strong quantitative, writing, and oral communication skills are also required. The strongest applicants will have demonstrated experience publishing in peer-reviewed journals, and with programming languages commonly used for statistical and scientific applications (e.g., R and Python), and familiarity with geospatial software (e.g., ArcMap, QGIS). Applicants should be physically fit, have significant field experience, be comfortable with flying in small planes and helicopters, and be capable of working in varied terrain for extensive periods in extreme conditions (e.g., temperatures that will range between 30 and -40 C, lots of biting insects etc.).

## Application instructions

Initially, all applications are to be sent as follows. Please email a cover letter with an explicit statement of analytical/quantitative AND field experience and abilities, current CV, unofficial transcripts, scientific writing sample and contact info for  $\geq 3$  references as a single attachment to [moose.project.applications@gmail.com](mailto:moose.project.applications@gmail.com). **The successful applicant is expected to begin in**

**spring or fall, 2027. Application deadline is November 15, 2026 but review of applications will begin immediately and continue until a suitable candidate is found. Once a successful applicant has been determined a formal application to the University is required, with a deadline of February 1, 2027.**

### Compensation

Students will receive a funding package of approximately \$34,000 per year. This includes a combination of a graduate teaching assistantship, internal fellowships and a research assistantship. Further, during the first 2-3 years of the project, students working fulltime in the field will be provided shared housing. Tuition fees can be found here:

<https://www.trentu.ca/graduatestudies/financial-matters/student-account-and-tuition/research-and-thesis-based-program-fees>. Numerous scholarships are also available and students are encouraged to apply (<https://www.trentu.ca/graduatestudies/tuition-awards-funding/graduate-scholarships-and-awards>). This funding package will not be reduced if a student is successful in obtaining a scholarship.

### Patterson and Northrup Labs

More information about the research groups of Drs. Brent Patterson and Joe Northrup can be found here: <http://www.canidungulatelab.wixsite.com/brentpatterson> and here:

<https://www.joenorthrup.com/>. More information on the project can be found here:

<https://www.ontariomooseproject.ca/>